

SATS — SIH Screening Round Improvement Plan

SIH26206 • Action plan based on the completed repository audit

Purpose: Ee document friend/developer ki direct ga implementation plan ga use cheyyadaniki. Existing app ni rebuild cheyyadam kaadu. Already strong ga unna parts ni touch cheyyakunda, screening lo expose ayye major gaps ni fix chesi, core disaster-response workflow ni genuinely working ga cheyyadam target.

Audit basis: Full static repository review, flutter analyze, flutter test (304 tests), and source-level verification. Runtime device execution was not performed in the audit, so final live demo must still be rehearsed on the actual device.

1. Overall Direction — Mundu em cheyyali?

Core idea ni discard cheyyakandi. App lo strongest part is disaster-focused citizen assistance + stealth SOS + location + geo-targeting + authority/responder workflow concept. Problem enti ante product chala broad ga undi, and key end-to-end links incomplete ga unnayi.

Simple ga cheppali ante: “Kotha features add cheyyadam” kanna, “already unna important features anni one complete disaster workflow ga connect cheyyadam” first priority.

2. Four Buckets — Change ni ila divide cheddham

Bucket	Meaning	Action
DO NOT TOUCH	Already strong / genuinely working	Only bug unte matrame modify cheyyali.
IMPROVE	Existing feature useful but incomplete/unsafe	Fix + verify + make demo-ready.
REMOVE	Dead, unrelated, risky or distracting	Safely remove/unlink; don't break core flow.
ADD — WORTH IT	New feature that directly strengthens core solution	Only high-value additions; no random features.

3. DO NOT TOUCH — Already Strong Parts

Stealth SOS native Android pipeline

Meaning: 731 lines Kotlin lo real work undi: SOS trigger, GPS, SMS, live tracking, foreground service. Idi current app lo strongest technical asset.

Action: Existing implementation ni unnecessary ga rewrite cheyyakandi.

3-tier GPS fallback

Meaning: Fresh location fail ayite last location, tarvata location updates fallback undi. Disaster context lo idi valuable.

Action: Existing implementation ni unnecessary ga rewrite cheyyakandi.

Boot persistence + hidden foreground service

Meaning: Service restart and stealth behavior already implemented. Unnecessary redesign cheyyakandi.

Action: Existing implementation ni unnecessary ga rewrite cheyyakandi.

MethodChannel bridge

Meaning: Flutter → Android native SOS chain end-to-end connected undi.

Action: Existing implementation ni unnecessary ga rewrite cheyyakandi.

Geo-matching

Meaning: Village → Mandal → District matching logic dedicated tests tho verified.

Action: Existing implementation ni unnecessary ga rewrite cheyyakandi.

Shelter validation

Meaning: Capacity/location validation tested and useful.

Action: Existing implementation ni unnecessary ga rewrite cheyyakandi.

Firestore role/incident immutability

Meaning: Privilege escalation and incident-field protection lo already good rules unnayi.

Action: Existing implementation ni unnecessary ga rewrite cheyyakandi.

Demo data labelling

Meaning: Demo data ni DemoTag tho honest ga identify chestunnaru. Keep it.

Action: Existing implementation ni unnecessary ga rewrite cheyyakandi.

4. IMPROVE — Existing Features ni Strong cheyyali

4.1 Authority / Responder Access

Problem: Current problem: Authority screens real CRUD tho unnayi kani real authority account provision cheyyadaniki proper flow/credentials levu. Meaning: app lo second half unna, screening lo show cheyyalemu.

Em cheyyali: Create a proper Admin SDK seeding script. At least one authority and one responder demo account provision cheyyali. Credentials/role/district/mandal/village README lo documented undali.

Login → dashboard live ga verify cheyyali.

Result: Expected result: Citizen incident/SOS submit chesina tarvata authority/responder side lo live workflow demonstrate cheyyagalali.

4.2 Responder SOS Visibility

Problem: Current problem: Emergency records initiator ki matrame readable. Responder app lo SOS feed ledu.

Em cheyyali: Firestore rule lo authorized authority/responder read access add cheyyali.

Authority/responder screen lo Active SOS list create cheyyali. Citizen, location, time, emergency status, assigned responder and resolution status show cheyyali.

Result: Expected result: “SOS vellindi — responder ki ekkada kanipistundi?” ane question ki live answer untundi.

4.3 Push Notifications

Problem: Current problem: FCM code/function references unnayi, kani firebase_messaging dependency/token flow complete kaadu. So push actually fire avvadu.

Em cheyyali: Two choices: (A) FCM fully implement: token save, refresh, Android channel, foreground/background handling, closed-app testing. OR (B) dead FCM code remove chesi SMS-first architecture ni honest ga document cheyyali.

Result: Expected result: App closed unna user ki alert vastundi ani cheppali ante actual device test pass avvali.

4.4 Firebase Configuration

Problem: Current problem: Android and web/iOS configuration different Firebase projects ni point chestunnayi; setup docs kuda stale.

Em cheyyali: One correct Firebase project select chesi firebase_options.dart, google-services.json, firebase.json, setup documentation consistent ga update cheyyali. flutterfire configure footgun remove/fix cheyyali.

Result: Expected result: Fresh setup chesina team member wrong project ki app ni accidentally point cheyyakunda run cheyyagalali.

4.5 Firestore Security

Problem: Current problem: Any authenticated user can create notifications for another user; users collection broad read access; complaint authorship/status restrictions weak; missing-person status weak.

Em cheyyali: Rules ni owner/role based ga restrict cheyyali. Notifications self-targeting or server-only creation. Users sensitive fields restrict. Complaints owner fields and editable fields restrict. Emergency authority/responder access intentional ga allow cheyyali.

Result: Expected result: Evaluator rules inspect chesina obvious security holes kanapadakudadhu.

4.6 Complaint Routing

Problem: Current problem: Complaint routing legacy 'admin' role ni query chestundi, while current roles are village_authority/district_authority/rescue.

Em cheyyali: Existing AppRoles.authorityRolesForQuery ni use cheyyali. Real authority notification/visibility verify cheyyali.

Result: Expected result: Citizen complaint success ani chupinchi backend lo orphan ga undakudadhu.

4.7 Loading / Empty / Error States

Problem: Current problem: Firestore response raanamundhe konni screens 'No data' show chestunnayi.

Em cheyyali: Every async screen: Loading → Data → Empty → Error states implement cheyyali. Network slow unna false empty message raakudadhu.

Result: Expected result: Live demo lo screen initial paint lo broken/empty laga kanipinchadu.

4.8 Profile Validation

Problem: Current problem: Validators pass chesina Form/validate flow complete ga use avvatledu. Invalid village/mandal data save ayye chance undi.

Em cheyyali: Proper Form + validate() add cheyyali. Save time lo actual error message chupinchali. Concurrent fields clobber kakunda update strategy correct cheyyali.

Result: Expected result: Geo matching break ayye invalid profile data prevent avutundi.

4.9 Correct SOS Notification Status

Problem: Current problem: guardiansNotified SMS attempt mundhe true ga write avutundi. SMS permission/number issue unna false success record create avvachu.

Em cheyyali: SMS loop complete ayyaka actual successful send count base chesi guardiansNotified/status set cheyyali.

Result: Expected result: Database status actual event ni represent chestundi.

4.10 Demo / Documentation

Problem: Current problem: Main README stock Flutter template la undi; large workflow document undi but evaluator-friendly setup ledu.

Em cheyyali: README lo purpose, problem, personas, architecture, setup, demo accounts, seed data, run commands, known limitations clear ga pettali.

Result: Expected result: Team member/evaluator ki project run cheyyadam and understand cheyyadam easy.

5. REMOVE — Unnecessary / Risky Parts

Legacy community-post / social features

Why remove: Disaster-management core ni dilute chestayi. Screening story broad ga kanipistundi. Important: Delete cheyyadam mundu dependency check cheyyali; core disaster workflow break avvakudadhu.

Unnecessary blood-bank / hospital modules

Why remove: Useful general civic features ayina, SIH26206 core disaster workflow lo primary kaavu. Demo time waste chestayi. Important: Delete cheyyadam mundu dependency check cheyyali; core disaster workflow break avvakudadhu.

Unreachable admin / my complaints / medical emergency screens

Why remove: Code undi kani reachable workflow lekapothe evaluator ki unfinished impression vastundi. Important: Delete cheyyadam mundu dependency check cheyyali; core disaster workflow break avvakudadhu.

Unused READ_SMS / RECEIVE_SMS / SYSTEM_ALERT_WINDOW permissions

Why remove: Supporting implementation lekapothe remove cheyyali; unnecessary permission questions avoid cheyyali. Important: Delete cheyyadam mundu dependency check cheyyali; core disaster workflow break avvakudadhu.

Dead FCM implementation — only if FCM is not being completed

Why remove: Half-working push code ni leave cheyyakandi. Implement or remove. Important: Delete cheyyadam mundu dependency check cheyyali; core disaster workflow break avvakudadhu.

Unused scripts and upload_hospitals.dart from lib/

Why remove: Release codebase lo tooling/seeding code mix kakudadhu.

Important: Delete cheyyadam mundu dependency check cheyyali; core disaster workflow break avvakudadhu.

Root build/debug logs and stale setup artifacts

Why remove: Repository clean ga maintain cheyyali.

Important: Delete cheyyadam mundu dependency check cheyyali; core disaster workflow break avvakudadhu.

6. ADD — Worthwhile Additions Only

Responder SOS Dashboard

Highest-value addition. Active SOS list + citizen + location + timestamp + status + assignment/resolution. Reason: SOS ni phone call ga matrame kaakunda actual response system ga prove chestundi.

Real Demo Accounts

Citizen + Authority + Responder. Reason: two-persona workflow live ga demonstrate cheyyadaniki mandatory.

Offline Incident Queue

Network lekapoyina incident local ga save → connection vachaka sync. Reason: disaster domain lo network failure direct real-world issue.

Firestore Emulator Security Tests

Citizen/authority/responder permissions and cross-user attacks test cheyyali. Reason: 304 tests mostly pure functions; runtime security proof strengthen avutundi.

Reliable Demo Fallback

Seeded dataset + 90-second recorded backup demo. Reason: venue Wi-Fi/device failure unna presentation continue cheyyagalali.

7. DO NOT ADD NOW — Time Waste Avoid Cheyyali

- Random AI/ML feature just for the word 'AI'. PS AI require cheyyadu; fake/weak AI value add cheyyadu.
- Extra dashboards without a real user need.
- More social/community features.
- More generic hospital/blood-bank modules.
- Decorative animations or complete UI redesign.
- Features that cannot be tested end-to-end.

Rule: Kotha feature add cheyyali ante, it must directly improve disaster response, reliability, security, or screening demo credibility.

8. Exact Priority Order

P0 — MUST FIX

1. Authority/responder accounts and access
2. Responder SOS visibility
3. Firebase project consistency
4. Firestore security holes
5. Correct guardiansNotified logic
6. Citizen → authority/responder end-to-end workflow

P1 — SHOULD FIX

7. FCM fully implement OR remove dead FCM
8. Loading/error/empty states
9. Complaint routing
10. Profile validation
11. README/demo setup
12. Release configuration

P2 — IF TIME ALLOWS

13. Offline incident queue
14. Accessibility labels
15. Working web fallback
16. Dead-code cleanup
17. UI polish

9. Final Target — App ela undali?

Target workflow:

Citizen
↓
Disaster alert / warning
↓
Geo-targeted information
↓
Incident report OR SOS
↓
Location captured
↓
Guardian / emergency contact notification
↓
Authority / responder receives incident
↓
Responder views location and takes action
↓
Status updated
↓
Recovery / follow-up

Critical rule: Demo lo claim chese every step actual ga work avvali. Mocked or unfinished step ni working ani present cheyyakandi.

Roman Telugu summary: “App ni pedda ga kanipinchela cheyyadam target kaadu. Manam already build chesina strong parts ni connect chesi, oka real disaster situation lo citizen → responder complete ga work ayye system ga prove cheyyali.”

10. Final Implementation Checklist

- ☐ Authority demo account login tested
- ☐ Responder SOS feed tested
- ☐ Citizen → incident → responder flow tested twice
- ☐ Citizen → SOS → guardian SMS tested
- ☐ FCM either fully working and device-tested OR removed
- ☐ Firebase project consistent everywhere
- ☐ Firestore security rules tested with allowed + denied cases
- ☐ Loading / empty / error states checked
- ☐ Profile validation checked
- ☐ Complaint routing checked
- ☐ Seeded disaster data confirmed in the correct Firebase project
- ☐ README updated
- ☐ Actual screening device tested
- ☐ Backup demo video ready

- ☐ No dead/unrelated module appears in the primary demo

Source note: This plan is derived from the completed SATS SIH screening audit provided by the team. The audit identified the native SOS subsystem, geo-matching, validation and security controls as strengths, and identified authority reachability, FCM, responder SOS visibility, Firebase split configuration, Firestore rule gaps, web build, loading states and scope dilution as major issues.